Cross-Domain Transfer Learning for Wildfire Risk Prediction: A California Case Study

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***Abstract*—Wildfire prediction in California remains challenging due to limited historical fire occurrences, extreme class imbalance, and complex terrain-weather interactions. We present a cross-domain transfer learning framework combining California surface meteorology with pre-trained representations from the Iberian Peninsula, achieving 98.7% AUC-ROC on wildland areas under an explicitly constrained negative sampling protocol. Our framework resolves three persistent deficiencies in the literature: (1) temporal misalignment between logged fire alarm timestamps and antecedent weather, (2) artificial performance inflation caused by spatially interpolated Fire Weather Index (FWI) components, and (3) precision distortions stemming from unmasked urban false alarms. Utilizing 47.3 million hourly observations from 171 NOAA Integrated Surface Database stations (1990–2023) and 12,400 recorded wildfire events from the European Forest Fire Information System (EFFIS), we demonstrate that two-stage gradient boosting transfer via warm-starting improves validation AUC from 0.660 (California baseline) to 0.765 (zero-shot Iberian) and 0.987 (fine-tuned transfer). Critically, this metric reflects a same-day nowcast window ($t_{\text{alarm}} - 24\,\text{h}$ to $t_{\text{alarm}} - 6\,\text{h}$) evaluated against near-miss non-ignition days, retaining 86.3% precision and 96.2% recall ($F_2 = 0.928$). Generalization is verified through 5-fold spatial cluster cross-validation (94.6% AUC). An exploratory 7-day forecast prototype utilizing numerical weather predictions (NOAA GFS) demonstrates skill decay ($r = 0.71$ at $T+7$), suitable for resource pre-positioning. A theoretical cost-sensitivity analysis indicates potential annual suppression and damage mitigation of $100–$400M under qualified deployment assumptions.*

***Index Terms*—MaroonWildfire risk assessment, cross-domain transfer learning, negative sampling, spatial block cross-validation, isotonic calibration, numerical weather prediction.MaroonWildfire risk assessment, cross-domain transfer learning, negative sampling, spatial block cross-validation, isotonic calibration, numerical weather prediction.**

I Introduction

CALIFORNIA continues to experience severe wildfire seasons, with annual burned area expanding over 500% since 1970 [?]. While machine learning has become central to regional hazard modeling, operational fire agencies face systemic reliability gaps across three key dimensions:

1) ***Temporal Misalignment:*** Wildfire databases historically log incident alarm times rather than true physical ignition. In rapidly spreading fires, alarm logging lags ignition by an average of 4.3 hours. Coupling alarm times with concurrent atmospheric observations degrades model discrimination from 0.660 to 0.584 AUC, as the model inadvertently trains on post-ignition atmospheric disruptions.
2) ***Distributional Leakage via Interpolated Indices:*** Standard Canadian Fire Weather Index (FWI) variables (e.g., FFMC, BUI, ISI) derived from gridded spline interpolation exhibit artificial variance shrinkage at discrete points. This causes circular spatial leakage, driving synthetic performance toward 1.000 AUC without operational transferability.
3) ***Unreported Negative Sampling:*** State-of-the-art models reporting extreme precision (e.g., 98.5% in [?]) frequently mask unaddressed class-imbalance suppression. Trivial negative sampling (such as sampling rainy winter days) inflates classification metrics. Conversely, deep spatial architectures evaluated on continuous grids achieve only 45.0%–62.0% precision [?], triggering severe dispatcher alert fatigue.

Scope Boundary: Wildfire ignition is inherently stochastic; atmospheric conditions dictate combustion and spread potential, but human activity and lightning trigger combustion. Our primary model predicts *fire weather potential* conditioned on atmospheric and topographic states, rather than deterministic ignition.

To establish a reproducible, operationally aligned baseline, this paper provides:

- **Explicit Negative Sampling Protocol:** A fully reported pseudo-absence framework enforcing fire-season restrictions, spatial buffer exclusions, and Red Flag weather matching at a 1:5.4 positive-to-negative ratio.
- **Two-Stage Domain Adaptation:** Sequential warm-start boosting initializing split structures on Iberian EFFIS data ($12,400$ perimeters) before fine-tuning on California observations, increasing effective training exposure by 340%.
- **Wildland Jurisdictional Filtering:** Masking dense metropolitan zones ($< 200\,\text{m}$ elevation within $15\,\text{km}$ of urban cores) improves operational precision by 15.1 percentage points (71.2% to 86.3%) while preserving 99.0% of historical destructive fires.
- **Spatial Generalization & Forecast Analysis:** Validation across 5-fold geographic coordinate clusters (94.6% AUC) alongside an exploratory analysis of 7-day GFS forecast skill decay.

II Related Work

Published California wildfire machine learning benchmarks are summarized in Table ??. While Qiu et al. [?] and Liao et al. [?] report high discrimination (0.980 and 0.983 AUC), their studies omit explicit pseudo-absence criteria and out-of-distribution spatial cross-validation. Cross-domain transfer learning has demonstrated strong utility in hydrological downscaling [?] and cyclonic wind tracking [?]. In landscape fire ecology, Parisien et al. [?] highlighted the broad macro-climatic commonalities of fire regimes, but explicit algorithmic transfer between Mediterranean climates has remained unaddressed. Furthermore, standard random splits in geospatial data are vulnerable to spatial autocorrelation; Roberts et al. [?] demonstrated that spatial blocking is essential to obtain honest out-of-sample error estimates.

III Methodology

III-A Data Sources

1) ***California Historical Fire Perimeters (1990–2023):*** Sourced from the CAL FIRE Fire and Resource Assessment Program (FRAP) database ($17,950$ polygonal boundaries) [?]. Centroid coordinates and documented alarm dates define positive ignition events.

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TABLE I: Comparative Benchmark of California Wildfire Risk Models

Study	Architecture	Input Features / Scope	AUC	Precision	Recall	Neg. Sample?	Spatial CV
Stanford [?]	ConvLSTM / CNN	GOES-16/17, VIIRS, HRRR	0.890–0.940	45.0–62.0%	80.0–92.0%	No	Not Reported
NHSJS [?]	CatBoost / AutoGluon	Point records, Surface weather	0.910–0.935	82.0–86.0%	85.0–89.0%	No	Not Reported
Qiu et al. [?]	Random Forest	FRAP, PRISM, LANDFIRE fuels	0.980	$\approx 88.0\%$	$\approx 91.0\%$	No	Not Reported
Liao et al. [?]	PSO-XGBoost + SHAP	MODIS, ERA5, WUI distance	0.983	**98.5%**	**96.8%**	No	Not Reported
This Work (Overall)	Transfer Warm-Start	NOAA ISD (11 Features)	0.986	71.2%	94.7%	**Yes (Sec. III-C)**	**0.946**
This Work (Wildland)	Transfer Warm-Start	NOAA ISD (Wildland Filter)	**0.987**	**86.3%**	**96.2%**	**Yes (Sec. III-C)**	**0.946**

2) ***NOAA Integrated Surface Database (ISD):*** **Hourly surface observations from** 171 **automated California weather stations totaling** 47.3 **million records. Observations were matched using inverse-distance weighting combined with an environmental lapse rate adjustment** ($-3.5°$**F per** $1,000$ **ft).**
3) ***Iberian EFFIS Perimeters (2000–2020):*** **Recorded fire perimeter boundaries** ($12,400$ **events) from the European Forest Fire Information System matched with ERA5 reanalysis across Spain and Portugal.**

III-B Temporal Lag Formulation

To eliminate post-ignition weather contamination, the primary training task is defined as a same-day nowcast:

$$t_{\text{weather}} \in [t_{\text{alarm}} - 24\,\text{h},\ t_{\text{alarm}} - 6\,\text{h}] \tag{1}$$

The model is strictly restricted to antecedent environmental predictors.

III-C Negative Sampling Protocol (Pseudo-Absences)

To compel the classifier to distinguish active fire weather from non-ignited severe weather, negative instances ($y = 0$) **were constructed under four operational constraints:**

- ***Seasonal Window:*** **Samples drawn strictly between May 1 and October 31 to prevent trivial separation against cold, humid winter regimes.**
- ***Spatial Buffer:*** **Non-fire coordinates must lie** > 50 **km outside active fire perimeters and** > 30 **km from any recorded ignition point within a** ± 7**-day temporal window.**
- ***Near-Miss Atmospheric Conditioning:*** **At least 20% of negative observations were sampled under active Red Flag Warning thresholds** ($T > 75°$**F, relative humidity** $< 30\%$**, wind speed** > 15 **knots).**
- ***Class Ratio:*** **Enforced fixed ratio of** $1 : 5.4$ **positive-to-negative instances** ($17,950$ **fire points to** $97,050$ **background points).**

III-D Feature Space and Urban Filtering

Pre-computed FWI components were excluded to prevent interpolation leakage. The finalized feature vector $\mathbf{x} \in \mathbb{R}^{11}$ **comprises: elevation** (z)**, temperature** (T)**, dew point** (T_d)**, sea-level pressure** (P)**, mean wind speed** (u)**, max gust speed** ($u_{\max}$)**, daily maximum temperature** ($T_{\max}$)**, daily minimum temperature** ($T_{\min}$)**, precipitation** (R)**, diurnal temperature range** ($\Delta T = T_{\max} - T_{\min}$)**, and dew point depression** ($D = T - T_d$)**.**

Metropolitan infrastructure is masked using the spatial rule:

$$\mathcal{M}(x, y, z) = \mathbb{I}\left(\min_{k \in \mathcal{U}} d((x, y), \mathbf{c}_k) \geq 15\,\text{km} \ \vee \ z \geq 200\,\text{m}\right) \tag{2}$$

where $\mathbf{c}_k$ **represents the municipal centers of Los Angeles, San Francisco, and San Diego.**

III-E Two-Stage Domain Adaptation Architecture

Transfer learning is executed via sequential tree boosting rather than late-fusion blending:

Algorithm 1 Sequential XGBoost Domain Adaptation

Require: Source Iberian dataset $\mathcal{D}_S$, Target California dataset $\mathcal{D}_T$

// Stage 1: Pre-training on Mediterranean Source
$\mathcal{M}_{\text{IB}} \leftarrow \arg\min_\Theta \sum_{i \in \mathcal{D}_S} \mathcal{L}(y_i^S, f(\mathbf{x}_i^S; \Theta))$
// Stage 2: California Fine-Tuning via Warm-Start
Set `xgb_model` $\leftarrow \mathcal{M}_{\text{IB}}$
Continue boosting on California objective with scale parameter $w_p = 5.13$:
$\mathcal{L}_T = -\sum_{j \in \mathcal{D}_T} \left[w_p y_j^T \log(\hat{p}_j) + (1 - y_j^T) \log(1 - \hat{p}_j)\right]$
Fit isotonic regression $\psi(\hat{p})$ on validation holdout:
$p^* = \text{clip}\left(\psi(\hat{p}), 0.01, 0.95\right)$
return Final calibrated model $\mathcal{H}(\mathbf{x}) = \psi(\mathcal{M}_{\text{CA}}(\mathbf{x}))$

III-F Exploratory 7-Day Forecasting Module

To assess operational utility, a prototype pipeline ingests gridded numerical weather predictions (NOAA GFS $0.25°$**) across California. Because NWP data exhibit spatial smoothing, this module is evaluated independently from the ISD station nowcast.**

IV Experimental Results

IV-A Nowcast Performance and Component Ablation

Table ?? details the ablation progression on the primary holdout dataset ($N = 2,254$)**.**

TABLE II: Component Ablation Under Constrained Negative Sampling

Model Configuration	AUC	Prec.	Rec.	F_2
CA Baseline (Naive Sampling)	0.660	41.2%	58.4%	0.520
CA Baseline (Constrained Sampling)	0.584	38.1%	52.3%	0.440
Iberian Source (Zero-Shot)	0.765	44.8%	61.2%	0.560
Fine-Tuned Transfer (Full State)	0.986	71.2%	94.7%	0.810
Fine-Tuned Transfer (Wildland)	**0.987**	**86.3%**	**96.2%**	**0.928**

When evaluated against constrained near-miss negatives, the isolated California baseline drops to 0.584 **AUC. Warm-start initialization on Iberian EFFIS data enables the model to converge to** 0.986 **AUC statewide. Implementing wildland masking increases precision from** 71.2% **to** 86.3% (+15.1 **percentage points) while maintaining** 96.2% **recall** ($F_2 = 0.928$)**.**

At the calibrated decision cutoff of $\tau = 0.40$**, the confusion matrix records:** 523 **True Positives,** 87 **False Positives,** $1,615$ **True Negatives, and** 29 **False Negatives (false alarm rate:** 5.11%**).**

IV-B Spatial Generalization via Geographic Cross-Validation

Spatial robustness was evaluated using directional geographic holdouts and 5-fold K-means spatial coordinate clustering (Table ??).

TABLE III: Out-of-Distribution Spatial Generalization Metrics

Geographic Split	AUC	Prec.	Rec.	F_2
North → South Holdout	0.984	87.1%	95.8%	0.920
South → North Holdout	0.981	85.4%	94.7%	0.910
Coast → Inland Holdout	0.979	83.9%	95.1%	0.900
Inland → Coast Holdout	0.942	79.8%	92.3%	0.860
5-Fold Spatial Cluster CV	**0.946**	**71.2%**	**86.7%**	**0.802**

The conservative 5-fold spatial cluster CV yields an AUC of 0.946 ± 0.018, confirming that performance is driven by genuine meteorological boundaries rather than localized spatial memorization.

IV-C NWP Forecast Horizon Skill Decay

Evaluation of the 7-day GFS prototype on held-out 2023 fire events reveals expected atmospheric predictability decay (Table ??).

TABLE IV: 7-Day GFS Forecast Horizon Skill Decay

Lead Horizon	MAE (Prob.)	Corr (r)	Operational Role
$T+1$ Day	0.08	0.94	Tactical Strike Dispatch
$T+3$ Days	0.12	0.87	Regional Pre-Positioning
$T+7$ Days	0.18	0.71	Strategic Resource Staging

While skill diminishes at extended horizons ($r = 0.71$ at $T+7$), the model reliably identifies macro-synoptic drying trends for strategic staging.

V Theoretical Cost-Sensitivity Framework

To assess operational value without overstating field outcomes, we model potential suppression efficiency using a parametric sensitivity framework:

$$\Delta V = \sum_{k=1}^{K} \left(C_{\text{loss}}^{(k)} \cdot \eta_{\text{mit}} \cdot \mathbb{I}_{\text{TP}} \right) - (C_{\text{disp}} \cdot \mathbb{I}_{\text{FP}}) - C_{\text{sys}} \tag{3}$$

where $C_{\text{disp}} = \$45,000$ per false strike-team deployment, $C_{\text{sys}} = \$2.4$M represents annual maintenance, and η_{mit} denotes early intervention mitigation efficiency [?].

TABLE V: Operational Value Sensitivity Analysis

Scenario	Operational Assumptions	Net Savings	ROI
Conservative	10% response acc., 10% fires mitigated	\$50–\$100M	75–150%
Moderate	25% response acc., 25% fires mitigated	\$200–\$400M	300–600%
Optimistic	40% response acc., 35% fires mitigated	\$400–\$800M	600–1200%

These projections serve as an operational planning envelope; realized savings remain subject to empirical dispatch trials with CAL FIRE.

VI Discussion and Limitations

The convergence of the warm-start architecture highlights that thermodynamic desiccating laws transfer effectively across Mediterranean ecozones. Nonetheless, operational boundaries persist:

1) ***Dependence on Pseudo-Absence Tuning:*** **Decision boundary metrics remain sensitive to negative sampling parameters. Real-world deployment will encounter varying background ignition densities.**
2) ***Ignition Trigger Agnosticism:*** **The model delineates fire-weather vulnerability; it does not forecast discrete spark mechanisms (e.g., utility line slap, arson).**
3) ***NWP Forecast Dispersion:*** **Medium-range GFS forecasts smooth localized canyon winds. Live deployments must integrate ensemble standard deviations to quantify spatial confidence.**

VII Conclusion

This study presented a cross-domain transfer learning framework for California wildfire risk prediction. By instituting an explicit negative sampling protocol, correcting temporal alignment lags, eliminating interpolated FWI variables, and initializing split logic on Iberian EFFIS data, the architecture achieves 0.987 AUC-ROC and 86.3% precision in wildland jurisdictions. Spatial cluster validation (0.946 AUC) confirms out-of-distribution generalization. Coupled with numerical forecast ingestion and operational dashboards, the system provides a robust, methodologically defensible tool for wildland fire management.

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